

A Comparative Study of AC-link and DC-link Based Frequency Converters

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Abstract

Frequency converters are widely used in variable speed drives, aircraft power supplies and wind energy applications. Traditional frequency converters are based on the DC-link topology. Large electrolytic capacitors are needed to eliminate the ripples in the DC stage. The size, lifetime and reliability of those capacitors are detrimental to system power density and robustness. The hard-switching loss of DC-link based frequency converter is also a considerable disadvantage of this topology. Recent studies of frequency converters are focusing on a new topology based on the AC-link. This new topology utilizes an LC resonating pair between the rectifier and the inverter. By controlling the charging and discharging time of the LC resonating pair, the converter is capable of delivering power in different frequency and magnitude. In addition, soft-switching can be obtained in the AC-link based frequency converter, which increases the system efficiency. The elimination of electrolytic capacitors also reduces the size of the converter which improves system power density and reliability. For comparison, both topologies are simulated in Matlab and implemented in hardware. A 45 kW prototype is designed, assembled, and tested in the lab. Experimental results are presented at the end of this paper.

Keywords: AC-link, DC-link, frequency converter, efficiency

I. Introduction

The electric grid provides power at 50 Hz and 60 Hz. An induction motor speed is proportional to the frequency of input power. To achieve variable speed control of the motor, a frequency converter is needed.

Aircraft power systems use 400 Hz to minimize motor size and increase power density. A frequency converter often acts as a ground unit to the aircraft to test system dynamic performance.

Wind turbines generate a wide range of frequencies in output power due to the unpredictable nature of wind energy. To prevent the pollution of power quality and sub-synchronous resonance (SSR) when connecting wind turbines to the grid, frequency converters can be implemented.

When connecting two systems with different frequencies, such as the industrial and residential applications mentioned above, DC-link frequency converters are commonly chosen. A DC-link frequency converter consists of a rectifier, a DC capacitor bank, and an inverter. Fig. 1 shows one implementation of this topology by using six pulse rectifier and IGBT-based voltage source inverter (VSI).

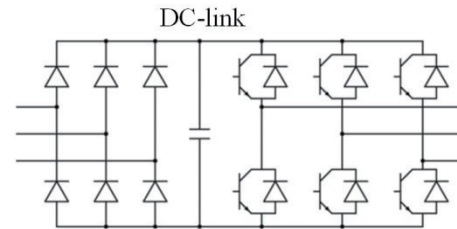


Fig.1. DC-link frequency converter

The six pulse rectifier will take three phase ac input voltage and rectifies it into DC voltage with considerable ripples as shown in Fig.2.

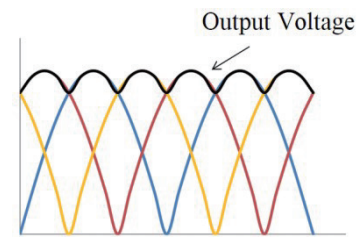


Fig.2. the output of 6 pulse rectifier

By choosing sufficient DC capacitance value, the ripple in DC stage can be greatly reduced. This capacitance value can be calculated using (1), where C is the link capacitance, V_{max} is the maximum ac line voltage, V_{ripple} is the minimum output voltage, R_{load} is the equivalent load resistance, and f is the ripple frequency. In a six pulse rectifier, f is six times the input frequency.

$$C = \frac{V_{max}}{V_{ripple} \times R_{load} \times f} \quad (1)$$

The VSI at the output needs a micro-controller to specify gate drive signals in order to generate output power at different frequencies. A three phase LC filter is also recommended to maintain sinusoidal output waveform and eliminate switching noise at high power levels.

The AC-link frequency converter introduces a LC pair between the rectifier and inverter. In order to create resonance in the LC pair, the rectifier and the inverter must have reverse blocking capability. It can be accomplished by using two IGBTs in series for each phase of the input and the output, while keeping the anti-paralleled diode in the IGBTs in reverse direction. Fig. 3 below shows the application of this topology.

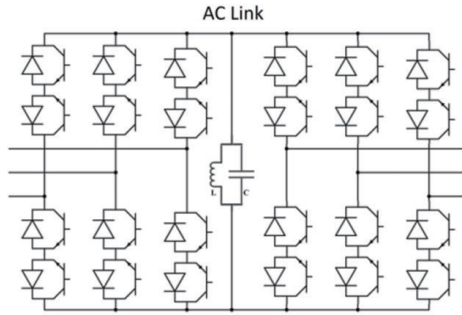


Fig.3. AC-link frequency converter

Alternatively, the reverse blocking ability can also be realized by putting a reversed diode at the collector of each IGBT, which are manufactured as reverse blocking IGBTs (RB-IGBTs).

II. The operation of AC-link frequency converters

The operation of AC-link frequency converters can be divided into 16 modes [2]. The first 8 modes allow energy to flow from input to output. The remaining 8 modes allow energy to flow from output to input. For simplicity, only first 8 modes are analyzed here. They are shown in Fig. 4 (a) and (b).

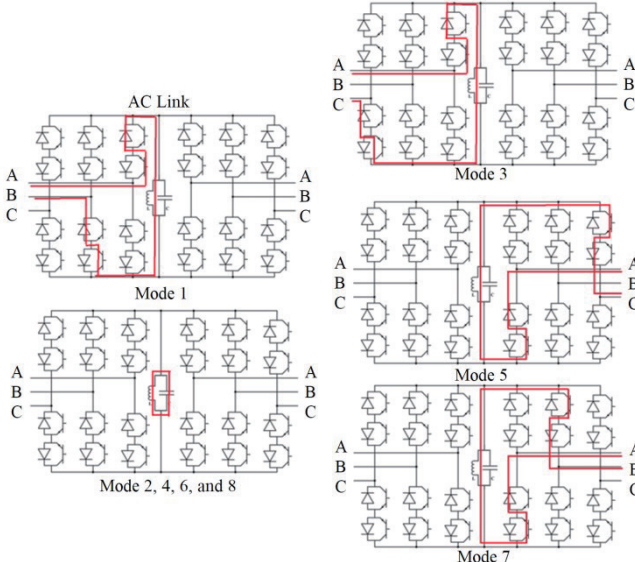


Fig. 4(a). Operation modes of AC-link.

Mode 1 (Charging): The LC pair is connected to the highest voltage phase pair V_{AB} . The link charges until input current reaches to the reference value. Then IGBT in phase B will turn off. Inductor current will start charging link capacitance.

Mode 2 (Partial Resonance): The IGBT in phase A will turn off. LC pair will resonate until the voltage of capacitor reaches second highest voltage pair V_{AC} .

Mode 3 (Charging): The link continues to charge until the current in phase A reaches the reference value. Then all IGBTs are turned off.

Mode 4 (Partial Resonance): The link inductor will continue to charge link capacitor until the link voltage is zero.

Mode 5 (Discharging): The IGBTs in output phase A and phase C will be turned on until the current in phase C reaches reference value.

Mode 6 (Partial Resonance): The link voltage will increase from V_{ACO} to V_{ABO} by LC resonance.

Mode 7 (Discharging): The IGBTs in phase A and Phase B will be turned on, allowing inductive current to discharge into phase B.

Mode 8 (Partial Resonance): The link voltage reaches negative maximum. The following modes 9-16 are similar to 1-8 except link voltage and current are negative values.

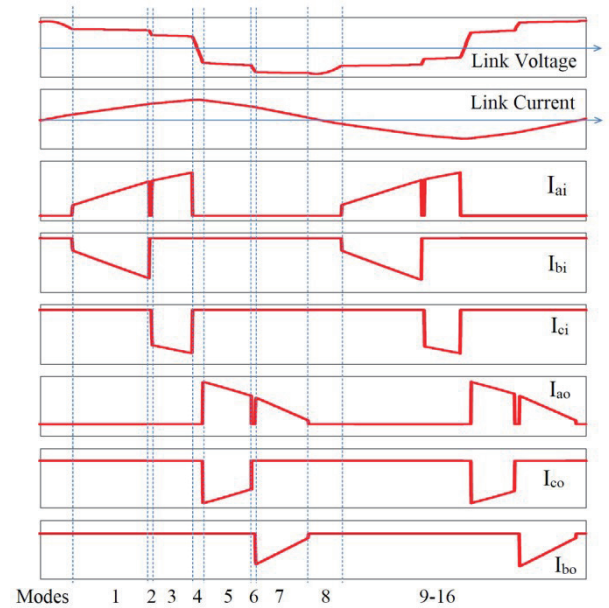


Fig. 4(b). Operation modes of AC-link in time domain.

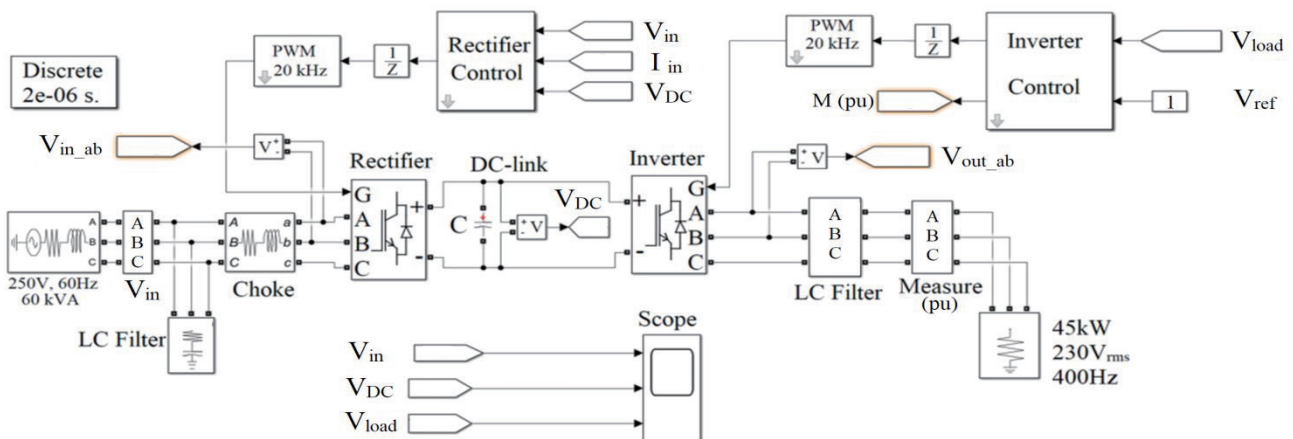


Fig.5 The Matlab model of DC-link frequency converter

III. Simulation Models

Both DC-link and AC-link frequency converters are simulated in Matlab. The simulation model for DC-link frequency converter is shown in Fig. 5 above. The input frequency is set at 60 Hz and the output frequency is 400 Hz.

The AC-link model is similar to Fig. 5 except capacitor C is replaced by an LC pair, and IGBTs in the rectifier and the inverter are using IGBT pairs in reverse directions. The gate drive signals are generated according to the 16 operation modes mentioned in section II. The link inductance is chosen using (2), where f is the link resonance frequency; P is the rated power [3].

$$L = \frac{P}{f \times I_{Link,peak}^2} \quad (2)$$

IV. Simulation Results

The DC-link simulation result is shown in Fig.6 (a). Three phase voltages are shown in both input and output. The source voltage V_{source} is measure at the output side of the grid transformer. It is rated at 250 V per phase and 60 kVA. The DC voltage V_{DC} is first raised to 850 V then drop to a stable 700V. The output voltage V_{load} is rated at 230 V per phase and 400Hz. The envelope of V_{load} is similar to V_{DC} during system initial transient. At steady state, V_{load} is close to one per unit (230 V) for all three phases.

The AC-link simulation result is shown in Fig. 6(b). The input frequency is 60Hz and output frequency is 400 Hz. The link frequency is set at 3.3 kHz.

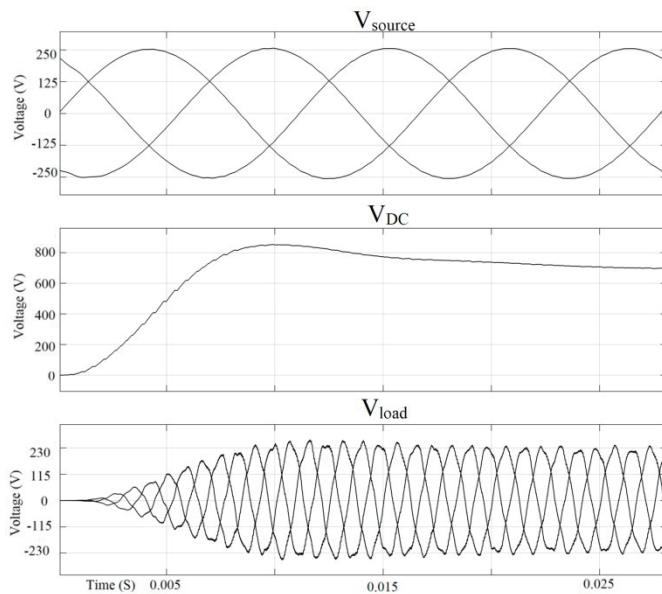


Fig.6 (a) The simulation result of DC-link frequency converter

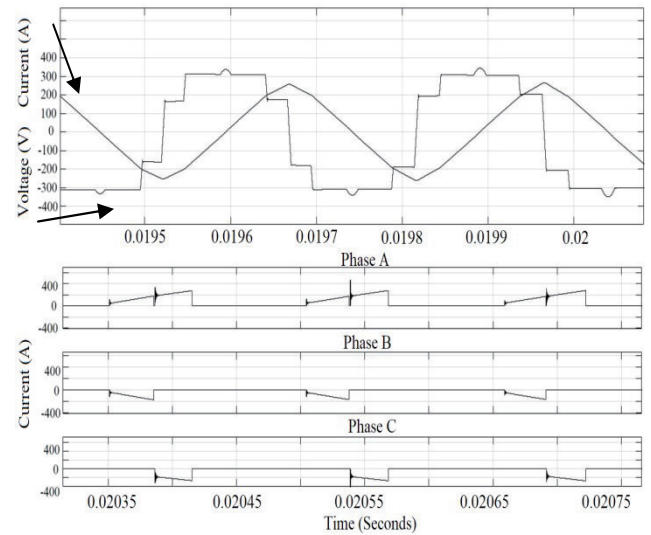


Fig. 6(b). The simulation result of AC-link frequency converter.

V. Experimental Verifications

A lab prototype has been designed, assembled and tested for both DC-link and AC-link frequency converters. The CAD drawing and the photo of this prototype are shown in Fig.7.

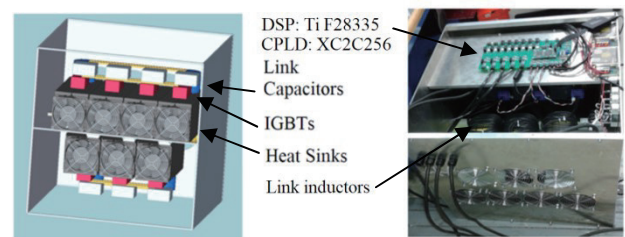


Fig.7 The CAD drawing and photo of the lab prototype.

The micro controller is TMS320F28335 from Texas Instruments. The sensing circuits and protections are processed by CPLD XC2C256 from Xilinx. The selections of electrical component in the prototype are shown in Table 1. There are 7 link capacitors in parallel. The inductors are made from toroidal core and litz wires to minimize core saturation and skin effect.

Table I. Electrical components of the prototype.

Component	Value	Part Number
IGBT	1200 V & 150 A	MG12150D
Link Capacitor	100 uF	944U101K122ACM
Snubber Cap	3 uF	SCD305K122C3Z25-F
Inductor	50 uH	0077101A7 (Core)

VI. Experimental Results

The link voltage and output voltage are shown in Fig. 8 below. The output frequency is set at 400 Hz. At maximum load, output power will reach 44.4 kW. The frequency converter is also capable of changing output frequency from 400 Hz to 800 Hz, a typical range for aircraft power systems.

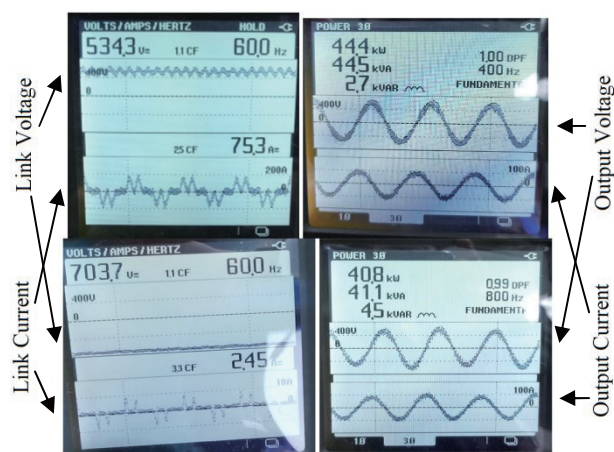


Fig. 8. Link and output voltage of the frequency converter.

VII. Comparison of AC-link and DC-link

At 400Hz and maximum load, the overall efficiency of the prototype when operating with a DC-link is 88.5%. The efficiency when using AC-link is 93.7%. The increase in efficiency mainly comes from the soft-switching of IGBTs and reduced link lost due to lower ESR of capacitors. However, this advantage also comes with a price. More IGBTs are needed. The control algorithm and sampling circuit are also more complicated, since the link resonating frequency is much higher than DC-link ripple frequency.

Another factor to consider when comparing DC-link and AC-link frequency converters is the EMI requirement. The LC resonating pair in AC-link can act as an antenna when frequency and power level are high. This will pollute the electromagnetic field distribution around the converter. Radiated EMI can induce faulty signals in the equipment, resulting a system shutdown or failure. A typical requirement for measuring the EMI performance is the DO-160 standard test. The test result of this frequency converter when operating with AC-link is shown below in Fig. 9. There are two spikes that are above the standard requirement, which make it unsuitable for aircraft system testing. EMI filter and Faraday cages can be added to the system to improve test performance. These are beyond the scope of this paper.

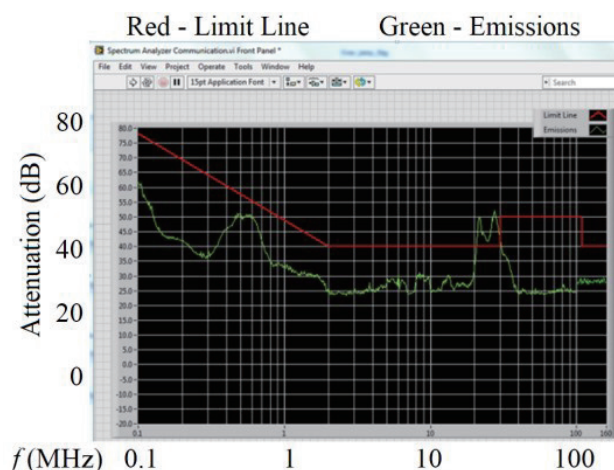


Fig. 9. DO-160 EMI test of AC-link frequency converter.

VIII. Conclusion

This paper compares the simulation and experimental results of DC-link and AC-link frequency converters. The DC-link simulation is built based on the six pulse rectifier and Voltage source inverter (VSI) with a DC capacitor bank. The AC-link simulation is built with IGBT pairs in reverse directions and an LC resonating pair.

A 45 kW lab prototype is built. It can operate in DC-link and AC-link mode. The output frequency range can be adjusted between 400 Hz and 800 Hz. The efficiency when operating in AC-link mode is 5.2% higher than that in DC-link mode. Other factors, such as power density, system robustness, IGBT total cost, control complexity and EMI requirements should also be considered when choosing between DC-link and AC-link frequency converters.

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